

STAA57-Winter 2024: Worksheet-2

Shahriar Shams

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Questions 1:

- i. Download the two csv files and load them to Rstudio. Lets name them “Book1” and “Book2” in R.
- ii. Merge the data sets using the column “location_id” and keep all the observations(rows) that belongs to Book2 and only the matching observations from Book1. Name the data set “m_d1”. (no need to print any data)
- iii. Create a subset of the data named Book2 and keep only those observations that DO NOT have a matching row in Book1 (matched using the column “location_id”). Name this data set “m_d2”. (no need to print any data)
- iv. Using data set “m_d1”, create a table which shows the sum of the column “UNOCCUPIED_ROOMS” by grouping them by the first two characters of the variable “LOCATION_POSTAL_CODE”. (hint: first create a variable that contains first two characters of the postal code and then use summarize with group_by). Use “na.rm =TRUE” option inside the sum to remove the missing observations while calculating the sum.

Questions 2:

- i. Create a scatter plot using the ggplot() command for “x=CAPACITY_ACTUAL_ROOM, y=UNOCCUPIED_ROOMS”. Use the LOCATION_CITY variable to define color and shape. Make sure to add a title (feel free to write anything) and change the x and y axis labels.
- ii. Once you have plotted, you will see that there are some points for which the location city is an empty character. What will be your suggestion in terms of showing them on the graph?

Instructions for completing this worksheet:

- Create a R markdown file, use your name and student id as the author name.
- Use the command ‘\pagebreak’ between your answers to Q1 and Q2 (this will make sure when you convert to pdf, answer to Q2 starts on a new page)
- Create an html output first, open it in any browser and then print or save as a pdf. (if you already know how to create a pdf directly from R studio then disregard this step)
- Upload your pdf to crowdmark in the place designated for Q1.
- Drag the pages corresponding to Q2 to the place designated to Q2.